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Paleomagnetism of the Late Precambrian Coldwell Complex, Ontario, Canada

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ABSTRACT

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The Coldwell Complex is a large multiphase alkaline intrusion on the north shore of Lake Superior. It was emplaced into Archean metavolcanics of the Superior Province in at least three distinct magmatic episodes during the Late Precambrian, giving isotopic ages ranging from 1010 to 1188 Ma. Detailed AF and thermal demagnetization was done on 416 specimens from 33 sites representing all petrologic phases. Additional data are included from an unpublished 1972 study by W.A. Robertson. The oldest magnetic component, PCA, was found in 19 episode I sites with a reversed mean direction of $D = 121.0°$, $I = -70.9°$ ($\kappa = 104$, $\alpha_{95} = 3.3°$). Its pole position falls on the Keweenawan APWP at an age of 1109 ± 5 Ma. The second component, PCB, was found in 11 sites with a normal mean of $D = 301.8°$, $I = 60.1°$ ($\kappa = 49$, $\alpha_{95} = 5.8°$) that indicates an age of 1103 ± 5 Ma. The youngest component, PCC, was found in nine episode III sites and gives a reversed mean of $D = 119.1°$, $I = -54.3°$ ($\kappa = 101$, $\alpha_{95} = 5.1°$) that indicates an age of 1095 ± 5 Ma. The assigned ages agree closely with recent U/Pb zircon ages but are distinctly older than the K/Ar and Rb/Sr ages. These paleomagnetic results require some changes in existing models for the petrologic evolution of the complex and indicate a much more rapid three-stage cooling history than previously postulated. They also provide strong evidence against the proposed concept of an asymmetrical geomagnetic field in Middle Keweenawan time.

Introduction

The Coldwell Complex is located on the north shore of Lake Superior (Fig. 1). It is an unusually large subcircular alkaline intrusion of about 500 km^2. It was emplaced into doubly folded greenschist to amphibolite-grade Archean metavolcanics of the Wawa Subprovince in the Superior Province of the Canadian Shield. The complex is bisected by Hwy 17 providing numerous road cuts for sampling (Fig. 2). It is Late Precambrian in age, giving Keweenawan isotopic ages ranging from 1010 Ma (K/Ar biotite; Currie, 1980) to 1188 ± 56 Ma (U/Pb zircon; Turek et al., 1985) that indicate either very slow cooling or analytical problems in one or more of the isotopic systems.

The cooling history of the Coldwell Complex has been debated. Turek et al. (1985) suggested that the complex was emplaced at about 1188 Ma with a subsequent cooling period in excess of 150 Ma. Heaman and Machado (1987) have suggested a much shorter cooling time of not more than 20 Ma after emplacement at 1108 Ma. At least three separate intrusive episodes have been recognized although the areal extent and order of each has been argued (Mitchell and Platt, 1978, 1982; Currie, 1980).

This study was undertaken on the premise that paleomagnetism might help solve some of the geologic and age problems with the Coldwell Complex. This seemed likely because the Keweenawan segment is the best defined part of the Precambrian apparent polar wander path (APWP) for North America (Halls and Pesonen 1982; Symons, 1989). In 1972 W.A. (Bill) Robertson had completed, but not published, a paleomagnetic study

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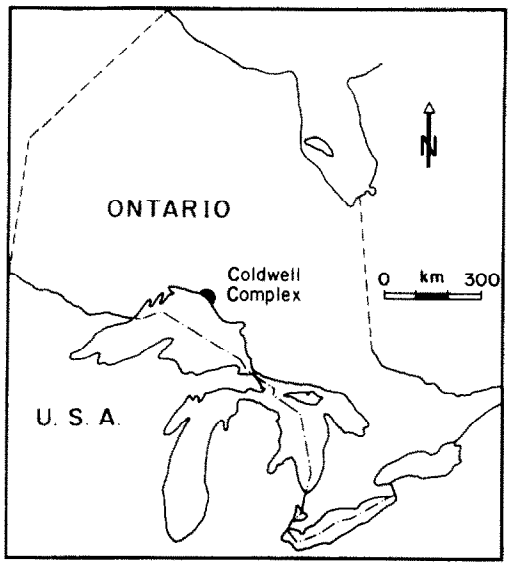


Fig. 1. Location of the Coldwell Complex in Ontario.

of the complex for much the same reason. With his consent we have included his table of results and used both data sets in arriving at our conclusions.

Geology

The geology of the Coldwell Complex is petrologically complex (Fig. 2; Puskas 1967). Both Mitchell and Platt (1978, 1982) and Currie (1980) have suggested that it was emplaced in three magmatic episodes that were separated by short time intervals as the magma source deepened and migrated to the present western side of the complex.

According to Mitchell and Platt (1982) episode I led to the emplacement of the eastern rim of gabbro, western zone of gabbro, and the extensive ferroaugite syenites (Fig. 2, 3a). After western migration of the magma source, episode II emplaced the ring of biotite gabbro and the nepheline syenite. Episode III emplaced the red syenites, quartz syenites and granites with large xenoliths of assimilated country rocks.

Currie (1980) included the eastern rim gabbro and ferroaugite syenite in episode I (Fig. 3b). All of the remaining rock units belonged to episode II except for the red syenites and granites, which he

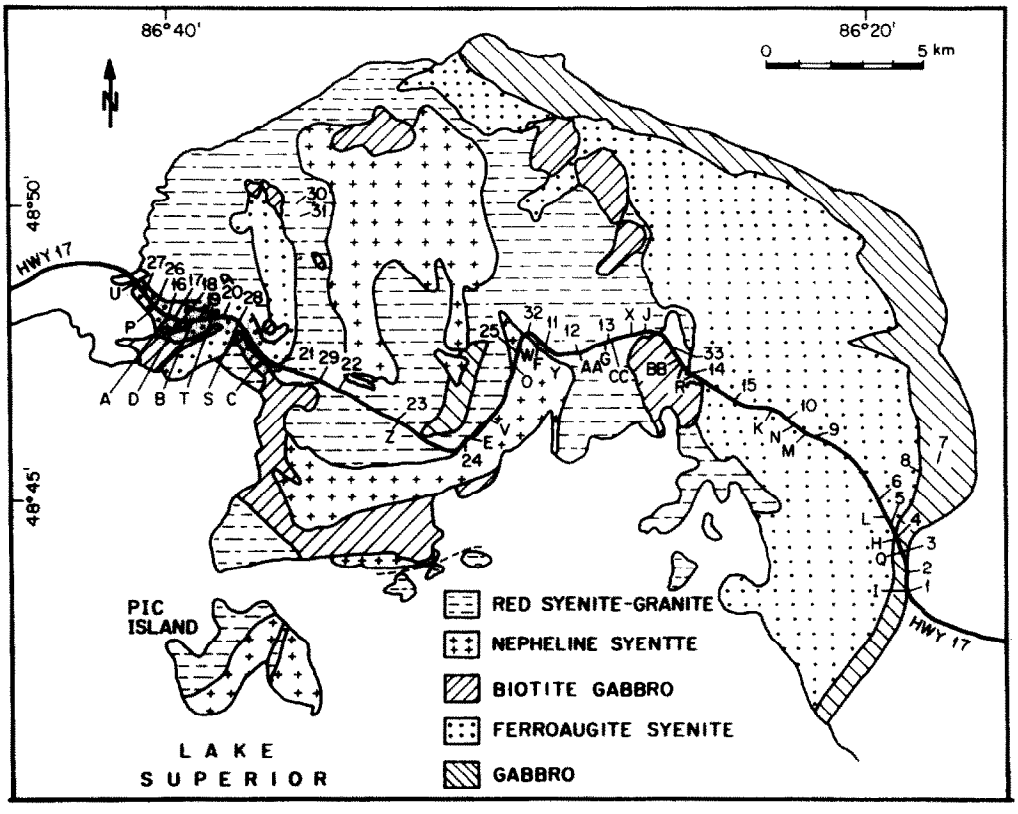


Fig. 2. Geology after Puskas (1967), Platt and Mitchell (1982) and Currie (1980), and site locations in the Coldwell Complex. The numbered sites are from this study; the alphabetic sites are from the unpublished data of W.A. Robertson.

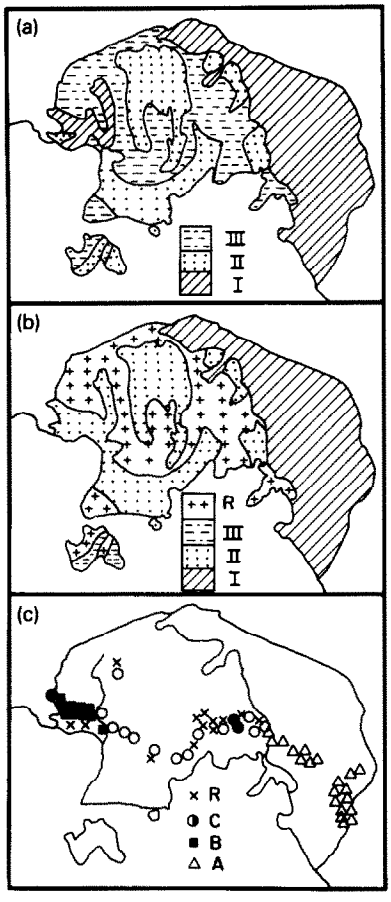


Fig. 3. Coldwell Complex maps showing: (a) the three intrusive episodes of Platt and Mitchell (1982); (b) the three intrusive episodes of Currie (1980); and (c) the distribution of A, B and C components found in this study and by W.A. Robertson. Site locations correspond to Fig. 2. *I*, *II*, *III* and *R* are the three episodes and metasomatized roof rock, respectively. *A*, *B*, *C* and *R* are the three magnetic components and the rejected sites, respectively, with normal (reversed) polarities shown as solid (open) symbols.

considered to be metasomatized roof rock and for the syenites on Pic Island which belonged to episode III.

Post-intrusive structures are limited to block faults with minor offsets associated with regional uplift (Mitchell and Platt, 1982).

Turek et al. (1985) have summarized the many isotopic age determinations done on the complex. Those with large error limits or with incomplete documentation are omitted here, and the following are corrected to the decay constants recommended by Steiger and Jäger (1977). Whole rock K/Ar dates of 1074 ± 39 and 1116 ± 41 Ma have been obtained by Stevens et al. (1982). Hurley (1958) obtained biotite K/Ar dates of 1075 and 1065 ± 50 Ma. A Rb/Sr whole rock isochron of 1045 ± 6 Ma has been reported by Platt and Mitchell (1982) and of 1070 ± 15 Ma by Bell and Blenkinsop (1980). Upper limit U/Pb ages of 1188 ± 56 Ma on zircon and of 1126 ± 6 Ma on tritomite from an episode II rock have been given by Turek et al. (1985). Recently Heaman and Machado (1987) obtained U/Pb zircon/baddeleyite ages of 1108 ± 1 Ma for episode I and II rocks of Mitchell and Platt (1982) with episode III about 10 Ma younger.

Sampling

The complex was sampled in roadcuts along Hwy 17 except for a few sites along side roads (Fig. 2). Five cores, oriented by sun compass, were taken at each of 25 sites (sites 1 to 25), and from one to three specimens were sliced from each core. At each of 8 sites (sites 26 to 33), four oriented hand samples were collected and two to four specimens were drilled from each. This gave a collection of 416 specimens from the 33 sites.

Measurement and orientation

The natural remanent magnetism (NRM) of each specimen was measured on an automated CTF cryogenic magnetometer inside a shielded room to minimize viscous remanent magnetization (VRM) effects from the present Earth's magnetic field (PEMF). Two pilot specimens from each site were alternating field (AF) demagnetized in 14 steps to 160 mT using a Sapphire Instruments (SI-4) AF demagnetizer. Two more pilot samples from each site were thermally demagnetized in 12 steps to 585°C using a Schonstedt TSD-1 thermal demagnetizer. The response of the AF and thermal pilots for each site was essentially the same (Figs. 4, 5). After the removal of an initial VRM in about the PEMF direction, a single component —either normal or reversed—was observed that decayed to the origin. Therefore the remaining specimens were AF demagnetized in three assigned fields of mostly > 20 mT by site to define their characteristic directions. All characteristic

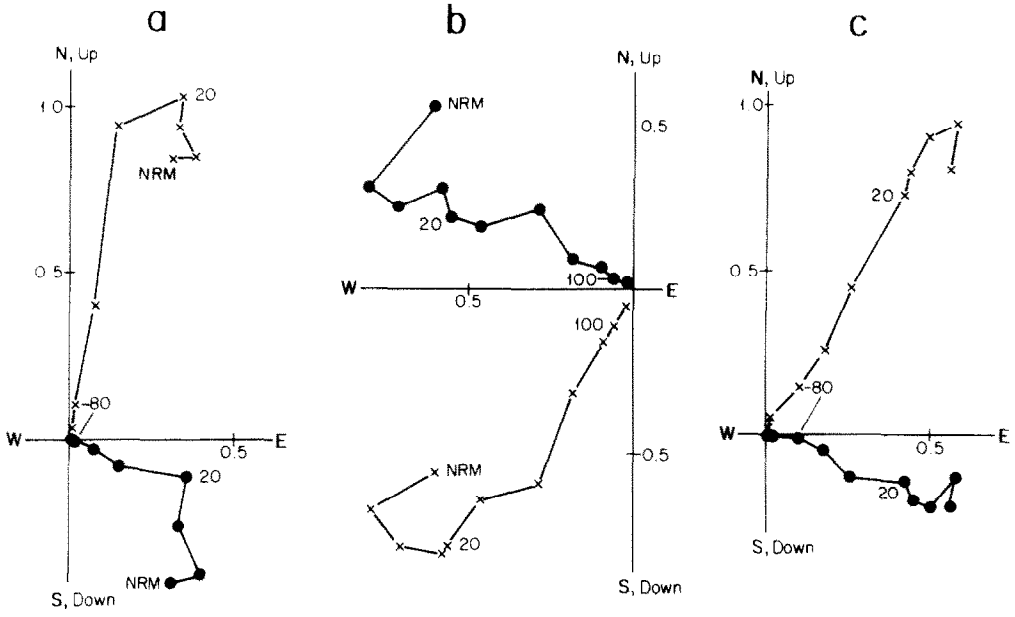


Fig. 4. Normalized orthogonal plots of example specimens on AF step demagnetization from: (a) site 1, A component; (b) site 17, B component; and (c) site 21, C component. Treatment is in milliteslas (mT); circles represent projections in the horizontal plane while crosses represent projections in the vertical plane. Axial intensities are expressed as a ratio of the NRM intensity.

directions have maximum angular deviations using principal component analysis of less than 10° with most less than 5° (Kirschvink, 1980).

Magnetic mineralogy

Isothermal remanent magnetization (IRM) testing was done on 15 representative specimens to better characterize their magnetic minerals. After AF demagnetization at 200 mT, each was magnetized in a direct current field in steps to 200 mT and then progressively AF demagnetized in six steps to 30 mT. All of the IRM acquisition curves were indicative of magnetite, and all the decay curves fall in the pseudo-single (PSD) to multidomain (MD) range of magnetite (Fig. 6). This im-

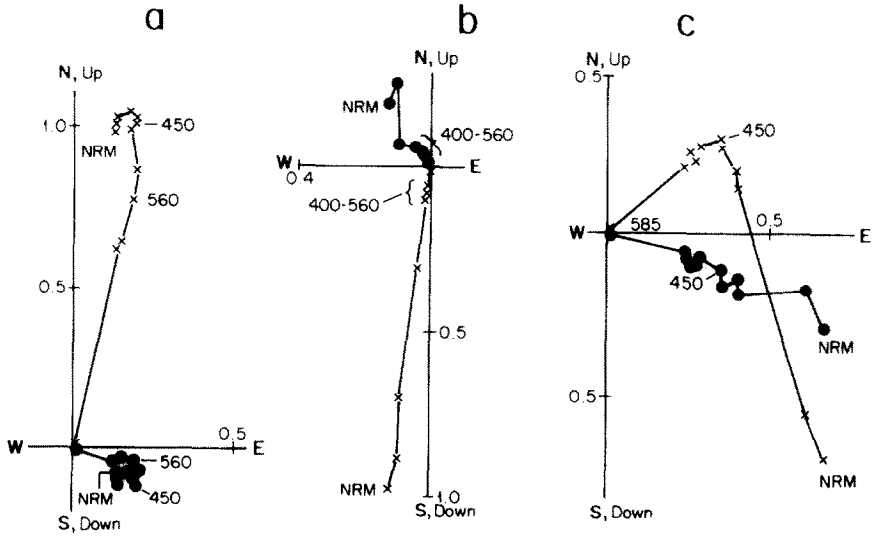


Fig. 5. Normalized orthogonal plots of example specimens on thermal step demagnetization from: (a) site 3, A component; (b) site 26, B component; and (c) site 22, C component. Treatment in degrees Celsius (°C). Plotting conventions as in Fig. 4.

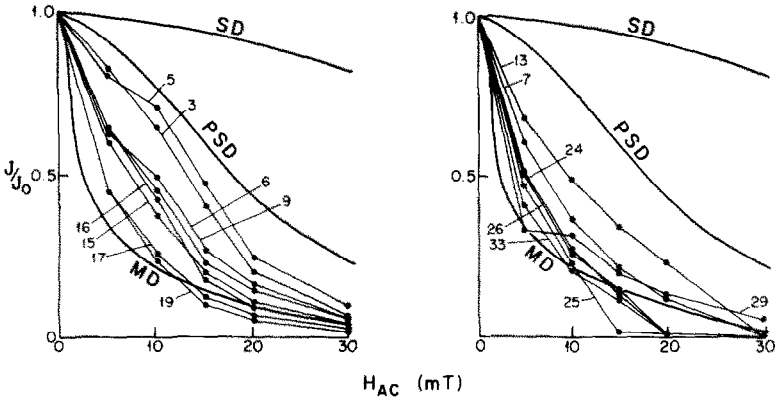


Fig. 6. Alternating field demagnetization of isothermal remanent magnetization in test specimens from the indicated sites. The type curves for single domain (SD), pseudo-single domain (PSD) and multidomain (MD) magnetite are from Dunlop (1972, 1973, and 1981).

plies that PSD magnetite carries the stable remanence, and this in turn agrees with the 550–585°C blocking temperatures found on thermal demagnetization (Fig. 5a, c).

Results and discussion

Twenty-seven sites gave a mean direction with an $\alpha_{95} < 30°$; however, two of the 27 acceptable sites (4 and 11) gave anomalous directions and were rejected (Table 1). Three of the other six sites were in red syenite–granite, two in the western zone gabbro and one in the nepheline syenite. Two of these latter six rejected sites (25 and 32) were near a minor fault which may account for their poor results. The 25 accepted sites means displayed three distinct remanence directions in three distinct geographic areas and are designated components A, B and C (Table 1, Figs. 3c, 7).

W.A. Robertson's unpublished data for the Coldwell Complex are listed in Table 2 and the site location shown in Fig. 2. His sampling strategy was similar to ours. After AF step demagnetization of one pilot specimen per site, he demagnetized the remaining specimens at one assigned AF field per site. Sites which we rejected as incoherent most likely reflect the inherent nature of the rock because five of his nine rejected sites (B,

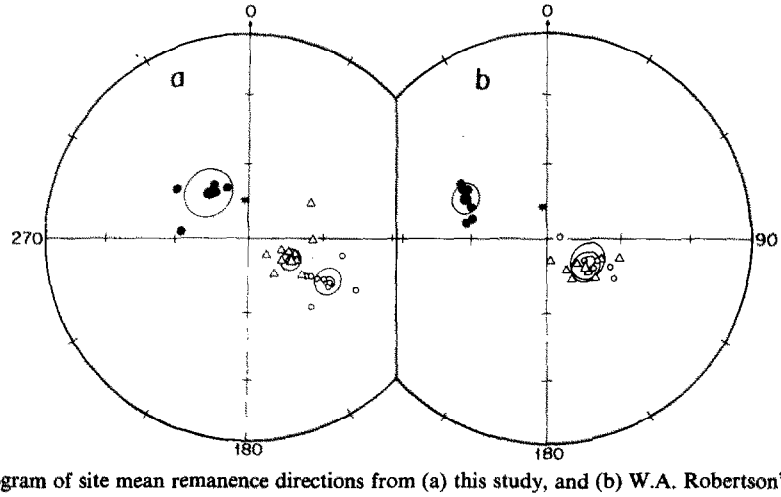


Fig. 7. Equal-area stereogram of site mean remanence directions from (a) this study, and (b) W.A. Robertson's study. Open triangles represent the A component, solid circles represent the B component and open circles represent the C component. Small symbols are individual site means; the larger symbols are component means which are circumscribed by their cones of 95% confidence after Fisher (1953). The asterisk indicates the present Earth's magnetic field direction.

TABLE 1

Site and unit means—this study

Site No.	Rock type	Demagnetizing level (°C/mT)	N	D (°)	I (°)	α_{95} (°)	κ
A component							
1	1	570/30, 60	15	110	−72	3	136
2	1	560/80, 125	14	125	−63	7	35
3	1	585/30, 80	10	130	−81	3	195
5	2	585/40, 60	17	92	−64	4	101
6	2	570/30, 40	12	124	−74	5	72
7	1	570/40, 80	10	110	−76	4	129
8	2	585/60, 80	6	115	−70	7	95
9	2	585/80, 100, 125	9	111	−71	5	117
10	2	570/30, 40, 60	6	117	−72	7	100
14	2	570, 60, 80	15	126	−63	3	236
15	2	570/10, 30	9	145	−71	5	114
Mean			(11)	117.8	−71.1	4.0	132
B component							
17	3	540/10, 15, 20	8	339	68	9	36
19	3	30, 40	5	329	64	17	20
20	3	570/20, 60	6	277	63	11	40
26	2	525/10, 20	10	328	68	9	27
27	2	570/20, 40, 60	8	306	55	13	18
Mean			(5)	313.9	65.3	11.3	47
C component							
13	6	540/20, 40, 60	13	300	57	2	366
21	7	570/60, 80	14	120	−59	3	228
22	7	560, 570/60	8	120	−53	4	174
23	7	560/40, 80	11	118	−56	5	98
24	4	585/20, 40	11	122	−62	4	151
28	2	585/20, 40, 80	10	119	−53	5	105
29	7	585/15, 30	10	116	−41	7	46
31	7	585/40	7	137	−52	16	16
33	3	570, 585/20, 40	15	102	−52	5	60
Mean			(9)	119.1	−54.3	5.1	101

Notes: N is the number of specimen endpoint directions for each site mean and (N) is the number of site means averaged for the component mean; the mean remanence direction is given by the declination (D) and inclination (I) in degrees; α_{95} is the radius of the cone of 95% confidence in degrees; and κ is the precision parameter of Fisher (1953). The rock types are: 1 = eastern gabbro; 2 = ferroaugite syenite; 3 = biotite gabbro; 4 = nepheline syenite; 5 = red syenite; 6 = quartz syenite; and 7 = granite. The mean for the C component was calculated using the antiparallel direction of site 13. The resultant vectors R, for A, B and C are 10.9241, 4.9142 and 8.9213, respectively.

F, G, P, and W) were adjacent to five of our six rejected sites (18, 32, 12, 16, and 25, respectively). The only normal polarity site that we found to the east of site C by the Little Pic River, is site 13 which is beside Robertson's sole normal site (site CC). Therefore we divided his data into the three geographic areas to try to duplicate components A, B and C.

A component

The A component was characteristic of 11 sites on the eastern side of the complex in units that

both Mitchell and Platt (1982) and Currie (1980) consider to be episode I (Fig. 3c). AF and thermal demagnetization showed that the A component is steeply reversed to the east-southeast and univectorial after the removal of a small VRM in the PEMF direction (Figs. 4a, 5a). The 11 sites give a mean direction of $D = 117.8°$, $I = -71.1°$ ($\kappa = 132$, $\alpha_{95} = 4.0°$; Table 1). Robertson's eight sites from the same area give a mean of $D = 125.3°$, $I = -70.6°$ ($\kappa = 76$, $\alpha_{95} = 6.4°$; Table 2) that is statistically the same at the 95% confidence level (McFadden and Lowes, 1981). The combined population gives a mean direction of $D = 121.0°$, $I = -70.9°$ ($\kappa = 104$, $\alpha_{95} = 3.3°$; Table 3). Its corresponding pole position, PCA, at 143.0°W, 54.2°N ($\delta_p = 5.0°$, $\delta_m = 5.7°$) falls directly on the Late Precambrian or Keweenawan segment of the APWP near the apex of the "Logan Loop" (Robertson and Fahrig, 1971) (Fig. 8).

PCA lies close to a number of other reversed poles from contemporaneous units including: (a) the Logan Sills (LS; Halls and Pesonen, 1982) dated at 1109^{+4}_{-2} Ma (U/Pb zircon; Davis and Sutcliffe, 1985); (b) the Logan Dikes (LD4; Pesonen, 1979) that are stratigraphically younger than the Logan Sills; (c) the lower Osler Group volcanics (OS3; Palmer, 1970) dated at 1108^{+4}_{-2} Ma (U/Pb zircon; Davis and Sutcliffe, 1985); and (d) the Marquette Dikes (MD; Pesonen and Halls, 1979) which are thought to be related to the

TABLE 2

Site and unit means—Robertson's study

Site No.	Rock types	H (mT)	N (N)	D (°)	I (°)	α_{95} (°)	κ
A component							
H	1	20	5	104	−59	4	465
I	1	20	7	127	−68	9	50
K	2	20	6	164	−81	18	17
L	2	40	7	145	−71	15	18
M	2	20	3	125	−74	13	85
N	2	20	3	143	−75	19	44
Q	1	10	5	108	−66	7	114
R	2	20	4	127	−64	10	81
Mean			(8)	125.3	−70.6	6.4	76
B component							
A	3	20	3	291	58	8	270
C	3	20	3	303	49	14	79
D	3	20	3	301	51	36	13
S	3	10	4	302	53	12	58
T	2	20	4	280	58	20	22
U	3	40	5	285	60	44	4
Mean			(6)	294.5	55.2	5.9	129
C component							
E	4	40	5	81	−85	29	8
J	4	20	5	113	−62	4	363
O	4	20	4	118	−73	13	64
AA	5	20	4	111	−68	5	426
BB	3	20	6	121	−68	9	57
CC	6	20	7	298	66	5	172
Mean			(6)	114.6	−70.5	7.1	91

Notes: Symbols as in Table 1 except H is the demagnetizing field in mT. The C component was calculated using the antiparallel direction of site CC. The resultant vectors R, for A, B and C are 7.9078, 5.9614 and 5.9449, respectively.

TABLE 3

Mean remanence statistics

Component	Mean remanence direction				Pole position			
	N	D (°)	I (°)	α_{95} (°)	long. (°W)	lat. (°N)	δ_p (°)	δ_m (°)
A—this study	11	117.8	−71.1	4.0	141.8	52.6	6.1	7.0
A—Robertson	8	125.3	−70.6	6.4	144.9	56.2	9.6	11.1
A—combined	19	121.0	−70.9	3.3	143.0	54.2	5.0	5.7
B—this study	5	313.9	65.3	11.3	161.5	59.6	14.8	18.3
B—Robertson	6	294.5	55.2	5.9	166.5	41.4	6.0	8.4
B—combined	11	301.8	60.1	5.8	164.9	49.1	7.5	10.0
C—this study	9	119.1	−54.3	5.1	170.4	43.3	5.0	7.1
C—Robertson	6	114.6	−70.5	71.1	142.3	50.6	10.6	12.3

Notes: N is the number of site means averaged for each component; the mean remanence direction is given by declination (D) and inclination (I) in degrees; α_{95} is the radius of the cone of 95% confidence in degrees; δ_p and δ_m are the radii of the ellipsoid of 95% confidence about the pole.

Logan Dikes. The PCA pole is also close to the normal Lackner Lake pole (LL; Symons, 1989) with a pooled whole-rock Rb/Sr isochron age of 1095 ± 30 Ma (Bell et al., 1982). The PCA pole position indicates that the A component is a primary untilted vector that records the direction of the Earth's magnetic field at the time of cooling at about 1109 ± 5 Ma. The error in the paleomagnetic age is obtained from the intersection points of the oval of 95% confidence with the APWP assuming a linear migration rate along the path. This magnetic age agrees well with the U/Pb zircon age of 1108 ± 1 Ma determined for episode I by Heaman and Machado (1987). Finally, the A

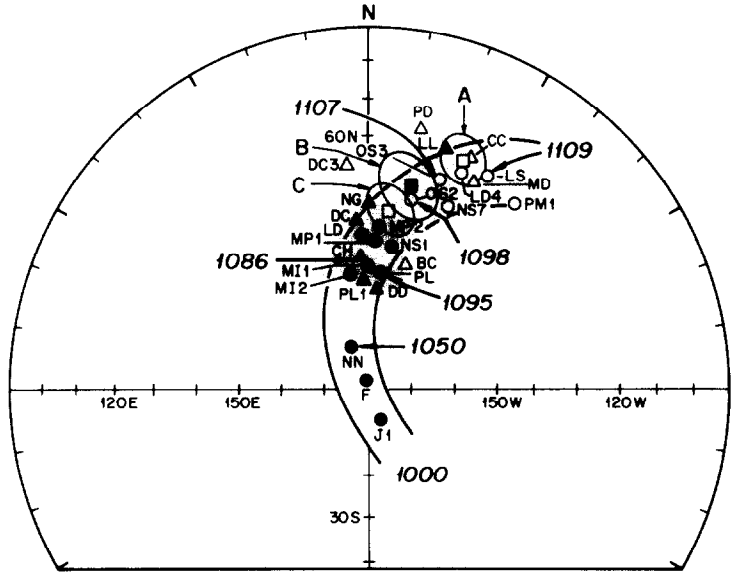


Fig. 8. Apparent polar wander path for Keweenawan time from Halls and Pesonen (1982) and Lewchuk and Symons (1988) showing the A, B and C component pole positions circumscribed by their ovals of 95% confidence. The added poles are for the Clay–Howells Complex (*CH*; Lewchuk and Symons, 1988), Nemegosenda Complex (*NG*; Symons and Garber, 1974) and Lackner Lake Complex (*LL*; Symons, 1989). The Powder Mill pole (*PM*) is from Palmer and Halls (1986).

component is found in episode I rocks only following either Mitchell and Platt's (1982) or Currie's (1980) model for the petrologic evolution of the Coldwell Complex.

B component

The B component was characteristic of five sites with normal polarity on the western side of the complex. AF and thermal demagnetization showed the initial removal of a minor VRM component in the PEMF direction, and then the emergence of a steep normal west-northwest univectorial component (Figs. 4b, 5b). The mean direction of $D = 313.9°$, $I = 65.3°$ ($\kappa = 47$, $\alpha_{95} = 11.3°$) for these five sites is marginally different at the 95% confidence level from the mean of $D = 194.5°$, $I = 55.2°$ ($\kappa = 129$, $\alpha_{95} = 5.9°$) for Robertson's six normal sites in the same area. Noting the small size of both populations, we think that they are recording the same magnetic signature and have opted to combine them to yield a mean of $D = 301.8°$, $I = 60.1°$ ($\kappa = 49$, $\alpha_{95} = 5.8°$). The resulting paleopole, PCB, at 164.9°W, 49.1°N ($\delta_p = 7.5°$, $\delta_m = 10.0°$) falls directly on the Keweenawan APWP also, but at a slightly younger age than the PCA pole. PCB falls between two reversed polarity poles from the upper and lower Osler Group volcanics (OS2, Halls, 1974; OS3, Palmer, 1970) which have given U/Pb zircon ages of 1097.6 ± 3.7 and 1108^{+4}_{-2} Ma, respectively (Davis and Sutcliffe, 1985). Thus we consider the B component to be primary and untilted, and to record a magnetization age of about 1103 ± 5 Ma. No rocks in this part of the complex have been dated using the U/Pb zircon method.

The differing polarities and poles for the A and B components argue strongly that the eastern and western border phases of gabbro and syenite do not belong to the same magmatic episode. We think that B records a second magmatic episode that is confined to the western side of the complex which occurred about 6 Ma after episode I with the A component. The B component was not found in any of the gabbro or nepheline syenite sites of episode II of either Mitchell and Platt (1982) or Currie (1980), and therefore intrusion of the western zone of gabbros with the B magnetization likely predates their episode II intrusive phases. Also Mitchell and Platt's (1982) inclusion of the western zone gabbros in episode I is not supported.

C component

We found the C component in nine sites in the central part of the complex. AF and thermal demagnetization characteristics are similar to those of the A component except that the initial VRM component is generally larger and that the inclinations are shallower (Figs. 4c, 5c). The mean direction for these sites is $D = 119.9°$, $I = -54.3°$ ($\kappa = 101$, $\alpha_{95} = 5.1°$), after converting the one normal site (13) to its antiparallel position. Robertson's six sites that correspond geographically, give a mean of $D = 114.6°$, $I = -70.5°$ ($\kappa = 91$, $\alpha_{95} = 7.1°$) that is different at the 95% confidence level. This difference cannot be explained by a systematic bias in either the collecting or laboratory procedures as this would have been found for the A and B components also. It may reflect small population sizes, particularly for Robertson's data which is biased to a steeper inclination by site E. It may also reflect differences in the demagnetization intensities, in the selection of characteristic directions, or in their screening for inclusion in the site means. Unfortunately, Robertson's individual specimen data are no longer available. Therefore, using only our data, the C component pole, PCC, at 170.4°W, 43.3°N ($\delta_p = 5.0°$, $\delta_m = 7.1°$) falls directly on the APWP to indicate an age of 1095 ± 5 Ma. It falls between the 1098 Ma OS2 pole and the two poles for the Mamainse Point volcanics (MP1 and MP2, Robertson, 1973; Palmer and Davis, 1987) that are directly overlain by a porphyry dated at 1086^{+1}_{-3} Ma (U/Pb zircon; Palmer and Davis, 1987). This age also agrees with the U/Pb zircon results of Heaman and Machado (1987) who found that the episode III rocks of Mitchell and Platt (1982) were emplaced at about 1098 Ma which include most of these sites. Thus the C component appears to be primary and to record the cooling of magmatic episode III in the Coldwell complex.

Several problems now appear for both petrologic models. One C component site (28) is in

ferroaugite syenite that both Mitchell and Platt (1982) and Currie (1980) assigned to episode I. The site could well have been remagnetized when the episode II or III rocks, depending on model, were intruded next to this site. If so, then it provides a contact test on a regional scale between episode I and the younger episode II or III rocks which suggests that they retain a primary remanence (Everitt and Clegg, 1962).

Six of the sites with the C component are in the red syenite to granite suite. This agrees with the Mitchell and Platt (1982) model who assigned these rocks to episode III. However, Currie (1980) considered them to be metasomatized roof rocks and so they should carry the magnetic signature of his episode II. However, since his episode II rocks carry both the normal B and reversed C components, his episode II category is suspect because the widespread distribution of the sites makes a bipolar remagnetization event improbable.

The remaining two C component sites (24 and 33) are in nepheline syenite. This agrees with Currie's (1980) model, assuming his episode II is in fact episode III in our paleomagnetic chronology. However, to fit the Mitchell and Platt (1982) model, these two C component sites would have to have been remagnetized in their episode III. This is possible because both sites are close to large bodies of granite. If they have been remagnetized, then they provide another contact test between episode II and III.

Our belief is that both Mitchell and Platt (1982) and Currie (1980) are wrong, and that the central part of the complex with the C component represents only one intrusive episode that we call episode III. It corresponds to Mitchell and Platt's (1982) episodes II and III and to Currie's (1980) episode II, so that Currie's episode III on Pic Island could represent a fourth episode.

Our episode III was emplaced during a period of dominantly reversed polarity although it must have spanned at least one polarity transition because site 13 is normally magnetized. The transition likely happened either very early or very late in the cooling sequence. If it was early then site 13 could record the transition from the normal B polarity period to a reversed C; however, if the transition was late then site 13 might record the onset of the prolonged Middle Keweenawan normal interval beginning at about 1090 Ma (Halls and Pesonen, 1982).

Reversal test

In addition to the unmetamorphosed nature of all of the rock types in the complex, to the absence of secondary overprints in any of the step demagnetized specimens excluding Holocene VRM, and to the possible contact tests noted above, the reversal test provides strong evidence that all of these rocks retain a primary remanence recording the age of intrusion. Following McFadden and Lowes (1981) and using only our data, the normal sites give a mean direction of $D = 314°$, $I = 65°$ ($N = 5$, $R = 4.9142$, $\kappa = 47$, $\alpha_{95} = 11.3°$); the reversed sites with the A and C components give $D = 119°$, $I = -64°$ ($N = 20$, $R = 19.6315$, $\kappa = 52$, $\alpha_{95} = 4.6°$); and the combined sites give $D = 122°$, $I = -64°$ ($N = 25$, $R = 24.5175$, $\kappa = 50$, $\alpha_{95} = 4.1°$). The ratio of precision parameters $r = \kappa_N/\kappa_R = 52/47 = 1.1$ is much less than the F statistic of 2.19 for 95% confidence, showing that both normal and reverse populations have similar distributions. Also the mean of each population, after reversal of the normal mean, lies within the other's cone of 95% confidence, showing that the normal and reverse populations pass the reversal test at the 95% level. This test indicates that primary remanences of the collection as a whole have been isolated that have not been biased by some secondary event. As noted below, the test works because the A and C components are about equally older and younger than the B component.

Asymmetry in the APWP

The ages assigned to the three components assume that the Keweenawan APWP records plate motion at a uniform rate and that the path is well-constrained temporally. The plate motion is almost entirely a latitudinal change and the rate is about 1°/Ma. This rate is fairly rapid but comparable to rates observed in, for example, the Indian continent during the Mesozoic or terrains in the North American Cordillera during the early Tertiary (Piper, 1987). Palmer (1970) found that

the Keweenawan Mamainse Point volcanics contained a reversed → normal → reversed → normal polarity sequence in which the reversed and normal poles on average are asymmetric by about 30°. He explained the result by postulating fault repetition of a reversed → normal sequence, coupled with imperfect removal of a secondary component. Robertson (1973) duplicated the Mamainse Point result and ascribed it to fault repetition, coupled with movement of the study area to lower paleolatitudes as plate motion produced the apparent polar wander path. Massey (1979) disputed the acceptance of fault repetition, finding geochemical differences between the "repeated" reversed and normal zones. From further study of the interbedded conglomerates, Palmer et al. (1981) found that they suffered a partial chemical remagnetization that could explain the asymmetry in the parental lavas but argued from contact tests that this was not the case. In their construction of the Keweenawan APWP, Halls and Pesonen (1982) opted for plate motion, and ascribed the reversed poles at the northern end of the path to a reversed superchron followed by a normal superchron on the descending track (Fig. 8, open and closed symbols). In contrast, Pesonen and Nevanlinna (1981) and Nevanlinna and Pesonen (1983) argued that the asymmetry was real and distorted the APWP and that the distorted path recorded the presence of either a long term offset in the Earth's dipole field or a non-dipole component. From 54 Keweenawan paleointensity determinations, Pesonen and Halls (1983) concluded that the paleointensities were consistent with plate motion from higher to lower paleolatitudes. Subsequently, Green et al. (1988) noted the asymmetry in the paleomagnetic data from Keweenawan dikes around Lake Superior and attributed it to a combination of apparent polar wander and of a non-dipole geomagnetic field. Recently Symons (1989) reported a normal pole for the Lackner Lake complex in the reversed superchron (*LL*, Fig. 8) and D.J. Dunlop (pers. commun., 1988) has found a reversed pole for the Shenango complex in the normal superchron. Both results require plate motion, and indicate more polarity transitions in the APWP than Halls and Pesonen (1982) defined.

The Coldwell Complex results provide evidence for apparent polar wander from plate motion by defining a reversed → normal → reversed sequence within one plutonic complex. The sequence tracks the APWP in the correct time series to place yet another normal chron in the reversed "superchron". In essence the Mamainse Point section, some 225 km southeast of the Coldwell Complex, provides the main evidence to date for an asymmetrical geomagnetic field in Keweenawan time. One possible contributor to this asymmetry at Mamainse is the incomplete averaging of secular variation. This is a common problem in flows because they erupt intermittently and cool rapidly (McElhinny and Merrill, 1975), and the two reversed zones at Mamainse Point are represented by only 5 and 15 flows (Palmer, 1970). Conversely, because of its size, the Coldwell Complex cooled slowly over 10^5 to 10^6 years to unquestionably average out secular variation (Carlslaw and Jaeger, 1959; Champion, 1980). It should be recalled that the upper reversed zone at Mamainse Point gives a paleolatitude between those of the normal zones above and below, and that it is statistically indistinguishable from the underlying normal zone after inversion of the direction (Palmer, 1970). Thus the paleomagnetic results for the section can be validly explained by plate motion combined with a reversed chron in the normal "superchron". Finally it is noteworthy that poles for the lower and upper reversed zones at Mamainse Point are only about 3° from, and statistically the same as, the reversed PCA and PCC of this study, respectively, and that PCA and PCC are statistically different from each other. Thus, even though the intervening normal zones are statistically different, it is tempting to correlate the two sections.

Cooling rates and burial depth

After the removal of VRM, the A, B and C components define their primary direction by about 400°C on the thermal demagnetization in the laboratory. This corresponds to a maximum burial temperature of about 200°C for single domain (SD) magnetite (Pullaiah et al., 1975: Walton, 1980). For PSD magnetite the burial tempera-

ture is even lower, perhaps down to about 140°C (Enkin and Dunlop, 1988). Using a geothermal gradient of 35°C/km for a rift zone (Morgan, 1983), the maximum 140°C burial temperature suggests that the burial depth upon intrusion was no greater than 4 km. This represents a conservative estimate because the geothermal gradient could be as high as 175°C/km in this sort of tectonic regime above active plutonism (Miyashiro, 1973).

Post-emplacement Keweenawan volcanism in the Lake Superior area, associated with the Mid-continental Rift System, led to the deposition of as much as 32 km of flows and sediments (Behrendt et al., 1988). The 4 km burial constraint for the Coldwell Complex implies that it was located at the margin of the rift zone where deposition was minimal. The acquisition of the A, B and C components within essentially one polarity period each, plus the absence of secondary Precambrian overprints, indicates that the magma for each episode cooled from the 585°C Curie temperature of magnetite to at least the 150°C burial temperature before the next episode was emplaced.

The 150 Ma cooling model for the complex of Turek et al. (1985) is not supported by these paleomagnetic results. Conversely the Heaman and Machado (1987) estimate of 20 Ma for emplacement is supported except that there were at least three cool-downs within no more than 20 Ma. Further the majority of the Rb/Sr isochron and mineral ages that fall in the 1030–1070 Ma range appear from the paleomagnetism to be too young to be cooling ages. Similarly the K/Ar ages which cluster about 1030 Ma (Turek et al., 1985) appear to be too young. It is hard to imagine any cooling model in which the three magnetization components are locked in by 1090 Ma while the Rb/Sr and K/Ar systems remain open for another 20–60 Ma, and this implies a problem in both the Rb/Sr and K/Ar systematics.

Conclusions

The rocks of the Coldwell Complex retain three untilted primary remanence components. Their definition supports in part the petrologic evolutionary models of Mitchell and Platt (1982) and Currie (1980) who both thought that the complex was formed in three magmatic pulses.

The oldest magnetization, the A component of episode I, is found in the gabbros and ferroaugite syenites on the eastern side of the complex. It records a concordant pole position with reversed polarity of about 1109 ± 5 Ma that agrees with the U/Pb zircon age of 1108 ± 1 Ma obtained by Heaman and Machado (1987).

The B component of episode II is found in the gabbros and nepheline syenites on the western side of the complex. It records a concordant pole with normal polarity of about 1103 ± 5 Ma.

The C component of episode III, is held in the biotite gabbros, nepheline syenites and red syenite to granite suite that comprise the youngest rocks in the center of the complex. Whether the rocks on Pic Island correlate with our episode III or represent a fourth episode remains open to investigation. The pole for the C component indicates an age of 1095 ± 5 Ma when the Earth's field was dominantly reversed. Its age also agrees with the U/Pb zircon age of about 1098 Ma obtained by Heaman and Machado (1987) for these rocks.

Because all three poles are concordant with the poles from contemporaneous Keweenawan units in the Lake Superior region, it is evident that the Coldwell Complex has not been substantially tilted or deformed since emplacement with the exception of up to about 4 km of simple uplift and unroofing.

The presence of a normal component chronologically between two reversed components is evidence for another polarity epoch in the Keweenawan APWP at about 1.1 Ga, and it indicates that the reversal record of the Earth's field for this time is more complex than originally thought by Halls and Pesonen (1982).

The poles for the reversed → normal → reversed sequence provide evidence for plate motion causing apparent polar wander during Keweenawan time rather than for the presence of an asymmetric magnetic field as postulated by Pesonen and Nevanlinna (1981), Nevanlinna and Pesonen (1983), and Green et al. (1988).

Based on the concordancy of poles for the two reversed intervals, it appears reasonable to tentatively correlate the Coldwell Complex polarity

zones with, $R_A \rightarrow N_B \rightarrow R_C$, with the lower three zones, $R_2 \rightarrow N_2 \rightarrow R_1$, found in the Mamainse Point section by Palmer (1970).

The univectorial behavior of all three components and the agreement of the paleomagnetic data with the U/Pb ages of Heaman and Machado (1987), suggests that each of the three intrusive episodes cooled rapidly from their emplacement temperature of about 850°C (Platt and Mitchell, 1982), to no more than about 140°C over a 5–7 Ma interval each within a total time interval of about 15–20 Ma.

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